Four infinite families of the Ahmed–Schaal conjecture on generalized Schur numbers

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Formal proof in Lean 4 with symbolic SAT certificates

Abstract. For $k \geq 3$ let L(k) be the equation $x_1+\cdots+x_{k-1} = x_k$. The generalized Schur number S(3; s,t,u) is the least N such that every 3-colouring of [1,N] has a solution of L(s) in colour 1, of L(t) in colour 2, or of L(u) in colour 3, in each case with every variable in that colour. Ahmed and Schaal (2016) conjectured that $S(3; s,t,u) = stu - tu - u - 1$ whenever $4 \leq s \leq t \leq u$. We prove the conjecture for $(s,t) \in \{(4,4), (4,5), (4,6), (5,5)\}$ and *every* $u \geq t$, which gives $S(3;4,4,u) = 11u-1$, $S(3;4,5,u) = 14u-1$, $S(3;4,6,u) = 17u-1$ and $S(3;5,5,u) = 19u-1$. The upper bound for all large u comes from one propositional formula per pair, in which the variables are indexed by linear expressions au+b rather than by integers. This extends the symbolic SAT method of Chang, De Loera and Wesley. The finitely many small u are handled directly. The whole theorem, including a general lower-bound construction, is checked in Lean 4 against a short statement file written from the definition. All 98 SAT certificates are LRAT proofs checked inside Lean and, separately, by an independent checker.

1. Introduction

Schur's theorem says that S(r) is finite: every r-colouring of [1,S(r)] has a monochromatic solution of x+y=z. Many generalizations replace x+y=z by the equation L(k) and allow a different equation in each colour. A solution may repeat values. For three colours and parameters $3 \leq s \leq t \leq u$, Ahmed and Schaal [1] proved the general lower bound $S(3;s,t,u) \geq stu - tu - u - 1$ and computed small values. They made two conjectures. Conjecture 1 concerns s = 3 and was settled in 2026 [2]. Conjecture 2 is the subject of this paper.

Conjecture 2 (Ahmed–Schaal [1]). For $4 \leq s \leq t \leq u$, $S(3; s,t,u) = stu - tu - u - 1$.

Before this work the conjecture had been verified only for finitely many triples by direct computer search. The tables in [1,2] list 16 triples with $s \geq 4$, all with $u \leq 10$. The best general upper bound, from [2], comes from the multicolour Ramsey number $R_3(s,t,u)$. It grows like u^{s+t-3} up to logarithmic factors, so it is far from linear in u. In the diagonal case the analogous identity $S(3;k,k,k) = k^3-k^2-k-1$ was proved for all k by Boza et al. [4], and later by a symbolic SAT computation [3].

Theorem 1. Let $M = st - t - 1$. For $(s,t) \in \{(4,4),(4,5),(4,6),(5,5)\}$ and every integer $u \geq t$, $S(3;s,t,u) = Mu - 1$. Thus Conjecture 2 holds for these four infinite families.

Note that $stu - tu - u - 1 = (st - t - 1)u - 1 = Mu - 1$, and that M is exactly the two-colour number S(2;s,t). The proof has three parts:

1. a lower-bound construction for every u (Section 2);
2. one symbolic formula per pair that gives the upper bound for all $u \geq u_0$ (Section 3);
3. direct certificates for $t \leq u < u_0$ (Section 4).

Section 5 describes the Lean formalization and Section 6 the computations.

2. The lower bound

The general lower bound is already in [1]. We give the following explicit form because it is the one formalized.

Lemma 2. Let χ_2 be a 2-colouring of $[1,M-1]$ with no colour-1 solution of L(s) and no colour-2 solution of L(t). For $u \geq t$, define a colouring χ of $[1, Mu-2]$ by $\chi(x) = \chi_2(x)$ for $x \leq M-1$, $\chi(x) = 3$ for $M \leq x \leq (u-1)M-1$, and $\chi(x) = \chi_2(x - (u-1)M + 1)$ for $x \geq (u-1)M$. Then χ has none of the three forbidden monochromatic solutions. Hence $S(3;s,t,u) \geq Mu - 1$.

Proof. Colour 3 is the middle block. Any $u-1$ summands from it have sum at least $(u-1)M$, which lies outside the block. Now take colour $i \in \{1,2\}$, with parameter $k \leq u$, and a solution with $k-1$ summands from the bottom block $[1,M-1]$ and the top block. There are four cases.

- If two or more summands are in the top block, the sum is at least $2(u-1)M > Mu-2$.
- If no summand is in the top block, the sum is at most $(k-1)(M-1) < (u-1)M$, so the target is in the bottom block. Then the solution is a χ_2-solution, which is impossible.
- If exactly one summand is in the top block, write it as $(u-1)M-1+y$. The sum is at least $(u-1)M$, so the target is in the top block too, say $(u-1)M-1+z$. Then $z = y +$ (the other summands), which is a χ_2-solution of L(k) in colour i. This is impossible. ∎

For each pair a suitable χ_2 was found by SAT. It is checked by evaluation in Lean (`smallColoringValid_st`). For example, for (4,4) the colouring of [1,10] is 2 2 1 1 1 1 1 1 2 2 up to symmetry.

3. The upper bound for all large u: a symbolic formula

Fix (s,t) and integers B, J. Let E be the set of pairs e = (a,b) with $0 \leq a \leq M$ and $-B \leq b \leq B$, excluding $a = 0$ with $b \leq 0$ and $a = M$ with $b \geq 0$. Read e as the linear function $e(u) = au + b$. Put $u_0 = \max(B+1, J+1, s, t, 4)$. For every $e \in E$ and $u \geq u_0$, $1 \leq e(u) \leq Mu - 1$. The formula F(s,t,B,J) has the following variables and clauses:

- **Colour variables** col[e][i], for $i \in \{1,2,3\}$, with exactly one colour per e.
- **Colours 1 and 2** (parameter k = s or t). Reachability variables $R_i[j][e]$, for $1 \leq j \leq k-1$, mean "e is a sum of j elements of colour i". The clauses are $\text{col}[e][i] \to R_i[1][e]$ and $R_i[j-1][e] \wedge \text{col}[f][i] \to R_i[j][e+f]$, where addition is coefficient-wise and $e+f \in E$. The forbidding clause is $\neg R_i[k-1][e] \vee \neg\text{col}[e][i]$.
- **Colour 3: c-padding.** The equation L(u) has a growing number of summands, so no fixed reachability depth suffices. For each constant c with $1 \leq c \leq \min(B,M)$, the variable $P_c[j][x]$ means "x(u) = (u−1−j)c + (a sum of j colour-3 elements)". The clauses are $\text{col}[(0,c)][3] \to P_c[0]$

[(c,−c)] and $P_c[j-1][x] \wedge col[y][3] \rightarrow P_c[j][x - (0,c) + y]$, for $1 \le j \le J$. Each step replaces one of the u−1 copies of c by y. The forbidding clause is $\neg P_c[j][x] \vee \neg col[x][3]$.

Lemma 3 (soundness; Lean `symForces_of_check_forall`). If F(s,t,B,J) is unsatisfiable, then $S(3;s,t,U) \le MU - 1$ for every $U \ge u_0$.

Proof. Let $U \ge u_0$, and let χ be a colouring of [1,MU−1] with no forbidden solution. Set col[e][i] true if and only if $\chi(e(U)) = i$. Set R and P to the true representability predicates at U. Every implication clause holds by construction. A forbidding clause can fail only if it exhibits a monochromatic solution; for colour 3 that solution has exactly u−1 summands, since $U-1-j \ge 0$. Two expressions may take the same value at U, but this is harmless, because the formula only gets weaker when it treats them as distinct. So χ induces a satisfying assignment, a contradiction. ∎

The formula does not depend on u, so a single SAT call settles infinitely many cases. The idea of indexing variables by polynomials comes from [3], who used it for 3-variable Rado equations. What is new here is the c-padding device for equations whose number of summands grows with the parameter, together with the use of off-diagonal colour classes.

Table 1. Certified symbolic formulas (CaDiCaL 2, direct LRAT, then lrattrim).

(s,t)	*M*	*B*	*J*	u_0	*vars*	*clauses*	*solve (s)*	*trimmed LRAT*
(4,4)	11	20	2	21	18,900	1,996,991	5.2	2.6 MB
(4,5)	14	30	1	31	32,414	5,218,154	16.1	13.3 MB
(4,6)	17	30	2	31	64,232	16,004,257	35.1	5.5 MB
(5,5)	19	30	3	31	100,746	31,176,775	73.3	6.0 MB

For smaller B, and for smaller J at the same B, the formula is satisfiable. We swept $B \in \{6,10,15,20,30\}$ and $J \in \{1,2,3\}$. A satisfying assignment does not give a real colouring. It only shows that the relaxation is too weak.

4. The finitely many small u

For $t \le u < u_0$ we use the direct encoding on [1,N] with $N = Mu - 1$. It uses the same reachability scheme, with colour 3 at depth u−1. All 94 instances (17 + 26 + 25 + 26) are unsatisfiable. Their direct-LRAT proofs took 160 s of solver time in total and occupy 16.3 MiB compressed. Each of the 94 values also has a satisfying assignment at $N - 1$, which an independent checker validated. That check is redundant given Lemma 2, but it confirms the encoding.

5. Lean formalization

The trusted part is a 33-line statement file (SHA-256 `c6ed1e48…6f35`). It contains no SAT notions.

```
def MonoSol (N k : Nat) (χ : Nat → Fin 3) (i : Fin 3) : Prop :=
  ∃ l : List Nat, l.length = k − 1 ∧ (∀ x ∈ l, 1 ≤ x ∧ x ≤ N ∧ χ x = i) ∧
    1 ≤ l.sum ∧ l.sum ≤ N ∧ χ l.sum = i
def Forces (N s t u) := ∀ χ, MonoSol N s χ 0 ∨ MonoSol N t χ 1 ∨ MonoSol N u χ 2
def Avoidable (N s t u) := ∃ χ, ¬MonoSol N s χ 0 ∧ ¬MonoSol N t χ 1 ∧ ¬MonoSol N u χ 2
def SchurEq (N s t u) := Forces N s t u ∧ Avoidable (N − 1) s t u
def ConjFamily (s t) := ∀ u, t ≤ u → SchurEq ((s∗t − t − 1) ∗ u − 1) s t u
```

The final file proves `conj_4_4 : ConjFamily 4 4`, and likewise for (4,5), (4,6) and (5,5).

The formulas are generated by Lean functions. Soundness does not rely on reasoning about the generators. Instead, a pure Boolean validator decodes every variable and checks that each clause has one of the shapes in Section 3, including the coefficient identities and membership in E. The proposition "every clause is valid" is established by `native_decide`. Soundness is proved once, generically, for any CNF all of whose clauses are valid. Certificates are stored gzip-compressed and Base64-encoded and checked by Lean's built-in LRAT checker (`Std.Tactic.BVDecide.LRAT.check`). A faulty decompressor could only make a check fail, never pass. The finite cases follow the same pattern.

`#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`, plus the auxiliary axioms that `native_decide` generates, one per decided computation. These cover the validators, the certificate checks and the χ_2 checks. So the trusted base is Lean's kernel, its compiler, and the statement file. The full build takes about 50 minutes on one core, dominated by the 94 finite certificate checks (23 min) and the validity check for the 31M-clause (5,5) formula (12 min).

Independently of Lean, every trimmed proof was checked by a separately written LRAT checker. The (4,4) symbolic formula was also regenerated from the written design by a second implementation, in Python, without reading the original generator. It was again unsatisfiable.

6. Limitations and outlook

For (5,6) and (6,6), the c-padding relaxation remained satisfiable at B = 30 (and for (5,6) also at B = 40, J = 1). Larger runs were limited by memory and disk. We expect richer padding to extend the method: several padded constants at once, or summands that are themselves linear in u. A two-parameter formula with expressions polynomial in both t and u could in principle settle Conjecture 2 for all t at fixed s. We did not attempt this.

On novelty, we checked the arXiv literature through October 2026 and the 55 works citing [1] that Semantic Scholar lists. None proves Conjecture 2 for an infinite family. We cannot exclude unpublished work.

References

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